Practical Optimization: Incorporating Uncertainty

Risk and Uncertainty are different "unknown" measure

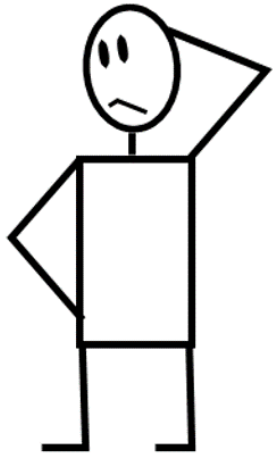
- Investors dislike both risk and uncertainty. Although both lead to uncertain outcomes, risk and uncertainty are different in nature.

- Risk is "Known unknown"
  - Must make a decision for which the outcome is not known with certainty
  - Can list all possible outcomes and assign probabilities to the outcomes

- Uncertainty is "Unknown unknown"
  - Cannot list all possible outcomes
  - Cannot assign probabilities to the outcomes

"Risk is measurable uncertainty"

"Uncertainty is unmeasurable risk"



The ideas for robust portfolio optimization

- Expected returns are difficult to forecast. The traditional portfolio optimization is usually overly reliant on expected return estimates and sensitive to small changes.

- Adding constraints can help to control the ill-effects of estimation uncertainty on optimization generated portfolio. However, this impact cannot always be mitigated by the presence of constraints.

- The uncertainty introduced by estimation error can be incorporated directly into the portfolio optimization process, referred to as robust optimization.

The approaches to robust optimization

Robust optimization refers to the technique of incorporating information about uncertainty set for the parameters in the optimization model. By using robust optimization, investors are likely to trade off the optimality of their portfolio construction against the risk of inaccurate estimation.

Classical robustness

- The fundamental for classical robust optimization is max-min preferences as insurance against worst case. This approach asks for the best performance when the least favorable scenario materializes.
- The big issue is to find the appropriate uncertainty set, which has impact on complexity of optimization and conservativeness of solution.

Regret robustness

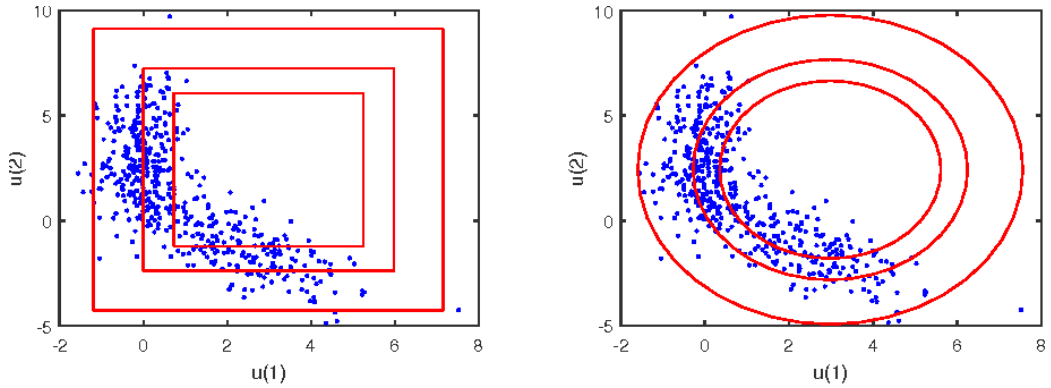
- The regret robustness takes all other possible performance rather than depends on solely on the performance under one special scenario, to minimize the “regret” of decision-making.
- The regret robust optimization is scenario-based, therefore, it may suffer the curse of dimensionality if scenario set is large.

Uncertainty set for classical robust optimization

- For classical robust optimization, the inputs for portfolio optimization are not only the traditional forecasts, such as expected returns $\hat{\mu}$, but also uncertainty sets including these point estimates.

- For interval uncertainty set, $U(\mu) = \{\mu | \, |\mu_i - \hat{\mu}_i| \leq \delta_i\}$, it can be interpreted as **individual confidence intervals** around each point estimate, such that each expected return is in the range $[\hat{\mu}_i - \delta_i, \hat{\mu}_i + \delta_i]$

- For ellipsoidal uncertainty set,$U(\mu) = \left\{\mu | \, (\mu - \hat{\mu})^T \Sigma_\mu^{-1} (\mu - \hat{\mu}) \leq \delta^2\right\}$, it is a **joint confidence region** that includes all vectors of expected returns within a certain number of standard deviations from the point estimate of the vector of expected returns.



The solution for classical robust optimization

- The RO applied to portfolio construction can be reformulated by modifying the MVO through a max-min process:
  - First, one finds, within the uncertainty set $\boldsymbol{U\mu}$, the worst-case expected returns of assets. They are defined as the realized returns that deviate most negatively from the estimated expected returns $\hat{\boldsymbol{\mu}}$.
  - Once the worst-case expected returns are obtained, the optimization process maximizes the portfolio returns, computed with the worst-case expected returns, under the risk constraint

$$\max_{\boldsymbol{w}}\left(\min_{\boldsymbol{\mu}\in \boldsymbol{U}_{\boldsymbol{\mu}}}\left(\boldsymbol{w}^T\boldsymbol{\mu}\right)-\frac{\lambda}{2}\boldsymbol{w}^T\boldsymbol{\Sigma}\boldsymbol{w}\right)\right)$$

The solution for classical robust optimization

- The max-min problem for robust optimization can be reformulated to a traditional portfolio optimization, depends on the choice of uncertainty set:
  - for interval uncertainty set, the equivalent objective function is (subject to same constraints),

$$Max \sum_i (\widehat{\mu_i} - \delta_i) w_i$$

which protects against the worst possible realization of each return forecast separately.

  - for ellipsoidal uncertainty set, the equivalent objective function is (subject to same constraints),

$$Max \sum_i \widehat{\mu_i} w_i - \delta \sqrt{w^T \Sigma_\mu w}$$

which protects against the worst possible joint deviation of the actual expected returns from the forecasts.

- For interval uncertainty set, the robust optimization model shrinks the expected return of assets with estimation error, which is similar as "**input hair-cut**" approach.

- For ellipsoidal uncertainty set, the additional term in the objective function as a "**risk-like penalty**" term that represents the joint deviation of the actual expected returns from the forecasts

The interpretation of classical robust optimization

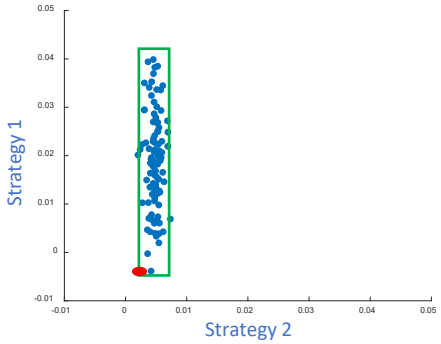
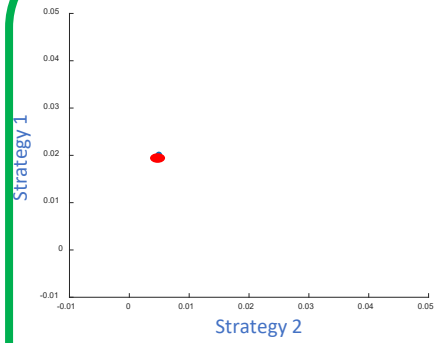
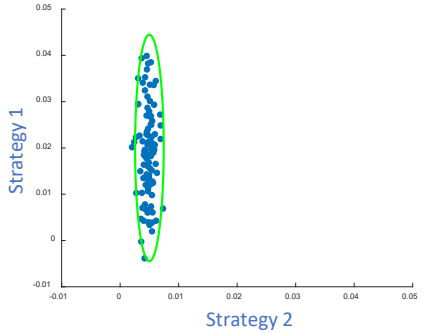
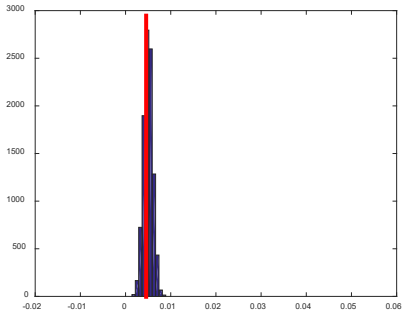
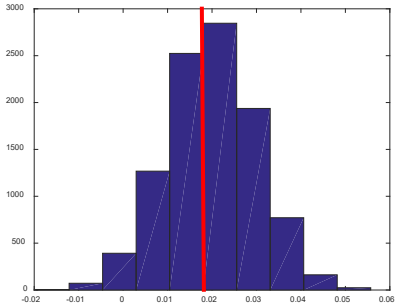
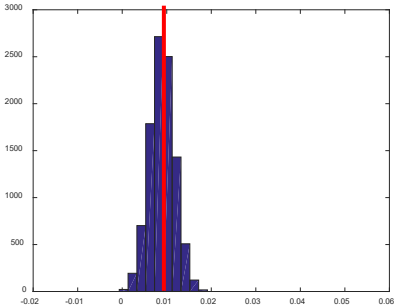
- For ellipsoidal uncertainty set, the equivalent objective function for robust optimization is (subject to same constraints)

$$Max \sum_i \widehat{\mu_i} w_i - \delta \sqrt{w^T \Sigma_\mu w}$$

- Some approximate methods for estimating the uncertainty matrix of forecasts Σ_μ assume proportional to the covariance matrix of returns. Under such a simplification, in the limit of large uncertainty in the estimation of expected returns, the robust optimization converges towards to risk-based allocation.

Proposed matrix for estimation uncertainty in expected returns	Limiting portfolio when uncertainty is large
Identity matrix	Equally weighted allocation
Diagonal matrix with estimated variance	Inverse variance allocation
Full estimated covariance matrix	Minimum variance allocation

An illustrative example for comparison of robust optimization

	Robust Opt- "EVA haircut"	Non-robust Opt	Robust Opt- "risk-like penalty"
Return Inputs			
Objective	$Max \sum_i (\widehat{\mu}_i - \delta_i) w_i$	$Max \sum_i \widehat{\mu}_i w_i$	$Max \sum_i \widehat{\mu}_i w_i - \delta \sqrt{w^T \Sigma_\mu w}$
Allocation	**W1=0%, W2=100%**	**W1=100%, W2=0%**	**W1=73%, W2=27%**
Total return distribution			

Strategy 1

Strategy 2

More conservative

Much less expected return, much more certainty

Less conservative

Less expected return, more certainty

- The performance testing shows that the "risk-like penalty" solutions have good overall performance, such as trade-off between the average scenario performance and the robustness of performance.

An illustrative example for MVO vs robust optimization

- We assume an identical Sharpe ratio that equals 0.46, the average of long term Sharpe ratio, for all assets. We prefer to use identical Sharpe ratio to show that MVO leads to large arbitrage positions in similar assets even when they have the same Sharpe ratio. For all portfolio optimizations, we impose the constraint that the volatility of optimal portfolio cannot exceed 10%.
- MVO generates a very counter-intuitive portfolio with large long-short positions in US Sovereign and US IG. The long-short position in two similar assets is particularly undesirable when one wants to define the strategic asset allocation. In contrast to MVO, RO yields a portfolio with no large long-short arbitrage positions, even without any constraints on weights.

	Volatility	MVO		RO with $\mathbf{\Omega} = \text{diag}(\mathbf{\Sigma})$	
		Weights	CTR	Weights	CTR
US Equity	19.14%	10.14%	1.45%	14.90%	2.32%
US Small Cap	23.70%	23.82%	4.23%	15.53%	2.75%
US Sovereign	9.89%	110.11%	8.15%	37.73%	2.77%
US IG	10.24%	-49.95%	-3.83%	25.24%	2.16%

Summary for classic robust optimization

- The scope of application for RO is extended considerably with the rise of robo-advisors in the finance industry. Indeed, to build automated robo-advisors, one needs a portfolio optimization algorithm that creates, without human intervention, well-balanced portfolios that are not highly sensitive to changes in expected return forecasts. From what we argued in this article, RO is clearly a better candidate for endorsing the role of portfolio optimizer in a robo-advisor than MVO.
- However, the calibration of uncertainty remains subjective and not solution fit-for-all purposes.

Utilizing Scenarios in Portfolio Construction

- Scenario-based allocation provides a systematic way to integrate the realistic economic performance scenarios and stress scenarios into the asset allocation process.

- Both investors' uncertainty of foreword-looking views and particular risk appetites can be factored into scenario-based asset allocation decisions.

Utilizing Scenarios in Portfolio Construction

	How	Why	
Foundations	• Structural economic and market themes • Cyclical economic and market roadmaps	• To understand and adapt to the current macro environment • To incorporate the future development of economic regime	Scenarios
Forecasts	• Create relevant macro scenarios • Assess probability for each scenario • Estimate market performance for each scenario	• To anticipate potential surprises in future market performance	Expected returns
Decisions	• Find optimal allocation in each scenario • Allocation decision takes account of all probable scenarios	• To mitigate downside risk and meet expectation • To help make robust decisions under conditions of uncertainty	Asset allocation

Building Macro Scenarios with Probabilities

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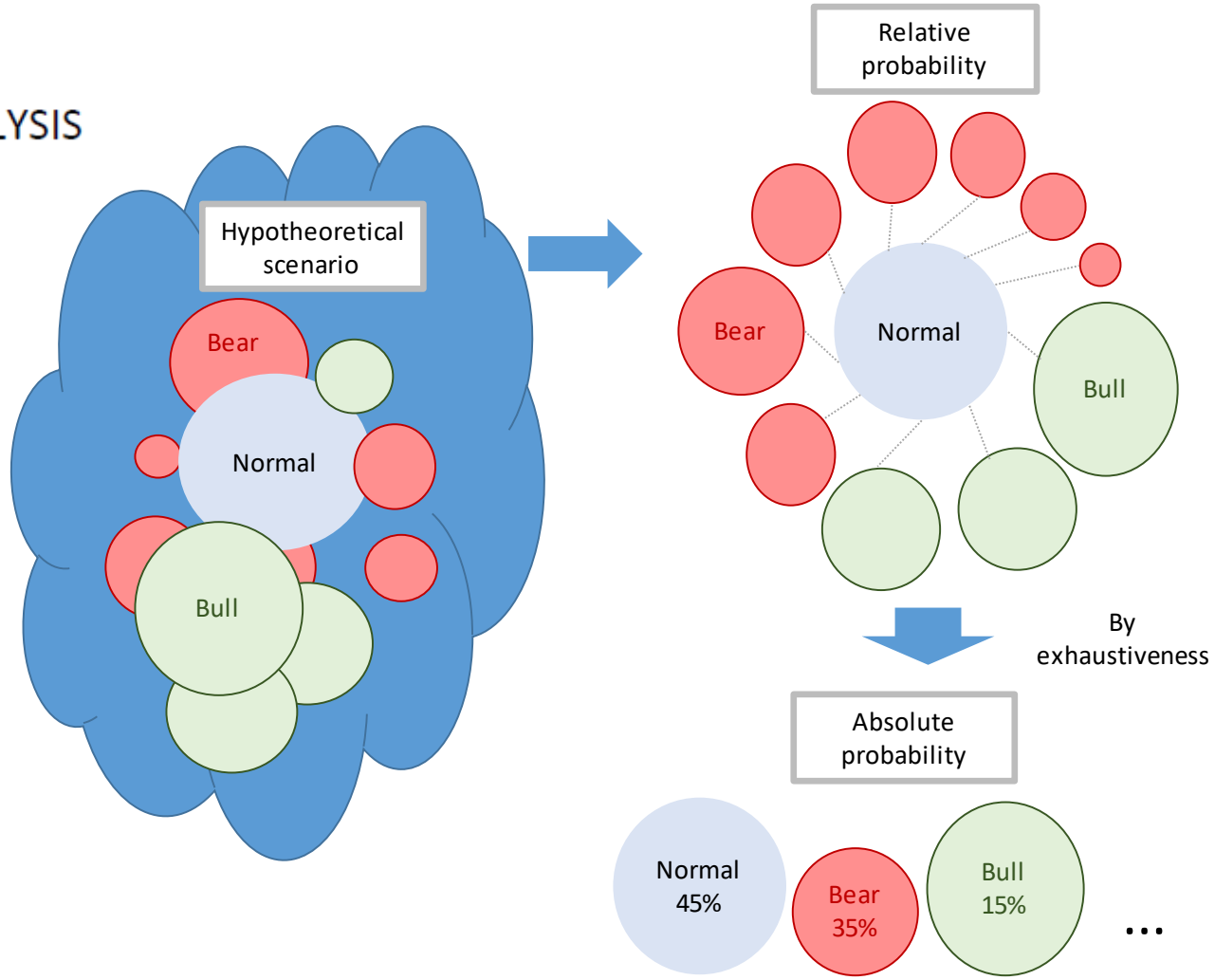
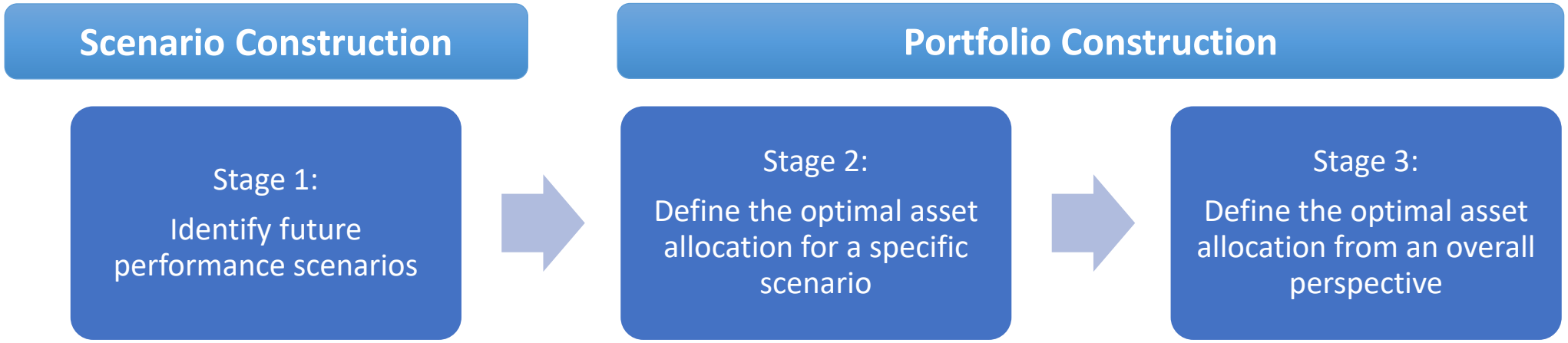


Robust Portfolio Construction with Scenarios



- Portfolio construction is a two-step process

- The last step is to aggregate all the results of scenario-specific asset allocation in order to define the optimum allocation that **meets all the investor's requirements and takes account of all plausible scenarios**.

Robust Portfolio Construction with Scenarios

- The term ‘regret’ in decision theory refers to the feeling associated with having decided on one alternative when it subsequently transpires that the choice of a different alternative would have resulted in a better outcome.

- The “**investment regret**” can be interpreted as the lost opportunity for return, and it arises because an investor does not know in advance which scenario will arise in the future.

Minimum-regret Optimization

$$minimize \sum_{S} \underbrace{P_s\left(R_{opt,S} - \vec{R}_s(w)\right)}^{2}$$

Scenario probability-weighted “investment regret”

- $R_{opt,S}$ denotes the optimal return for each scenario-specific optimal allocation
- $\vec{R}_s$ denotes the scenario dependent return vectors for overall optimal allocation
- P_s designates the probabilities of scenarios S

Robust Portfolio Construction with Scenarios

- Investment Objective: maximize the scenario probability-weighted investment regret, subject to risk target constraint

- The "investment regret" objective function is able to comply with the safety-oriented attitudes of investors
  - N=1, it represents linear distance measure, the optimal solution converges to the mean-variance approach (in AR context)
  - N=2, it represents quadratic distance measure, large deviations from the scenario-specific optimal return $R_{opt,S}$ are penalized more than smaller deviations

Robust Portfolio Construction with Scenarios

- We can also construct “scenario safe” portfolio that provides protection against pessimistic or stress scenarios by ensuring that the portfolio meets certain performance criteria.

- A simple approach is to incorporate the constraint requiring a minimum return to be achieved in each scenario into the optimization problem.

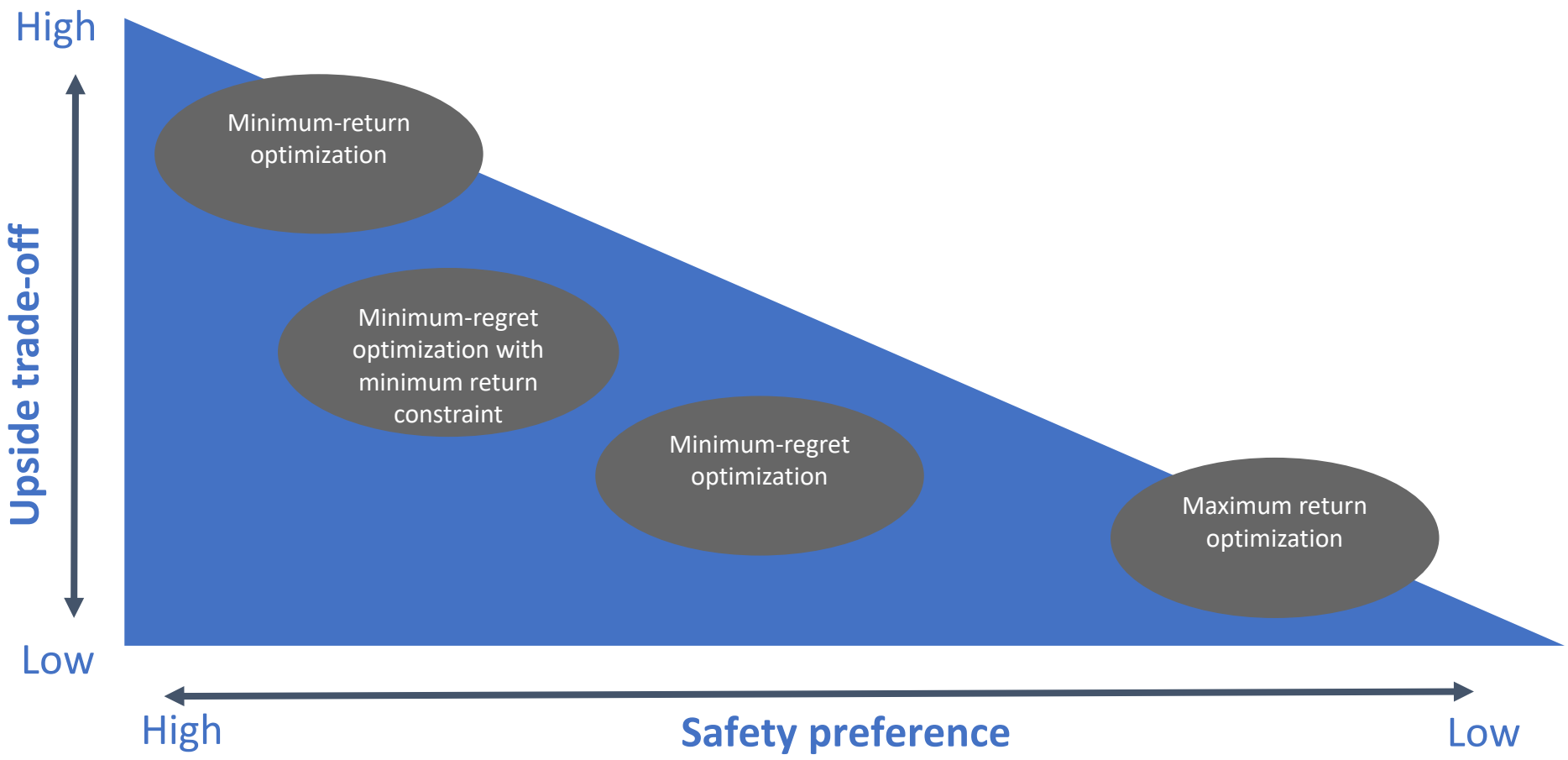
- Another way is Minimum-return optimization for investors with a strong safety focus.

Minimum-return Optimization

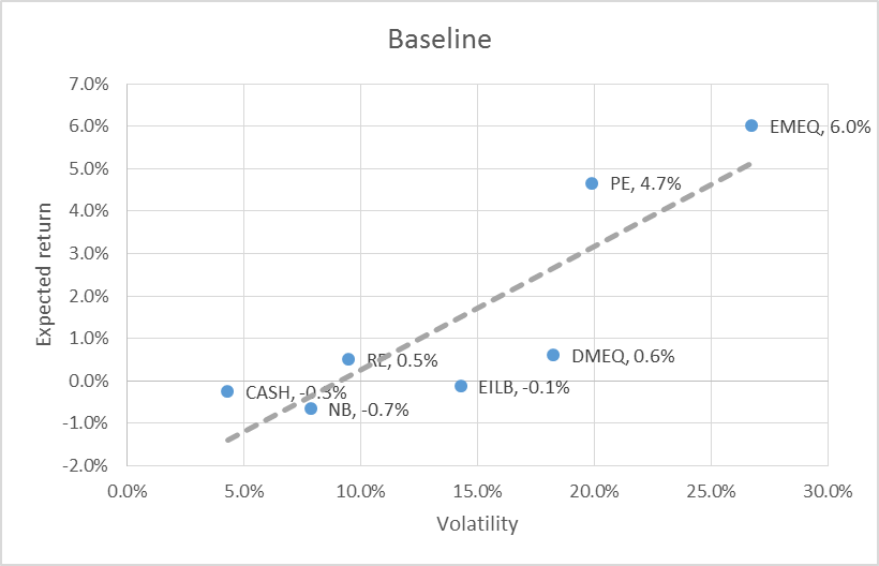
$$minimize \sum_{S} P_s\left(\vec{R}_s\vec{w} - minimum\ return\ target\right)^2$$

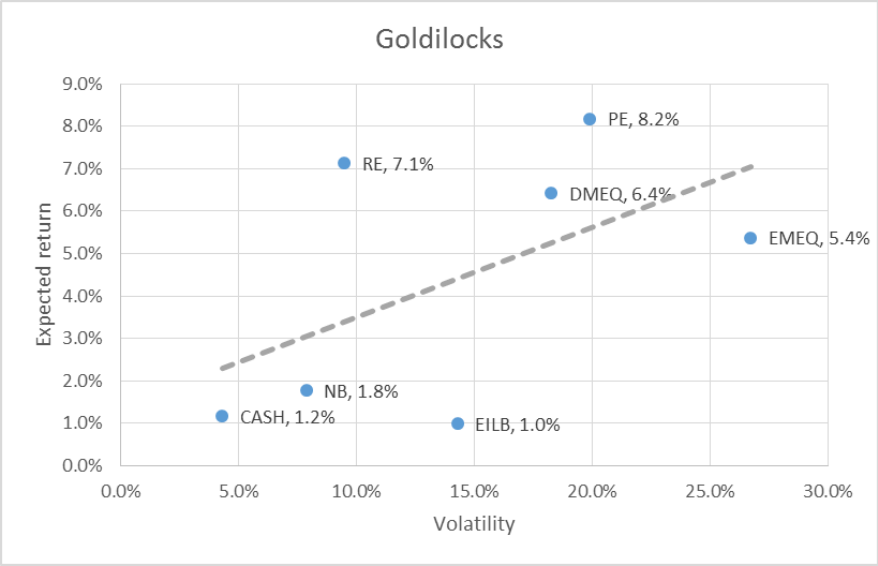
- Both portfolio construction approaches are safety-oriented and conservative, in return for which they are consciously forgoing any upside potential in some scenarios.

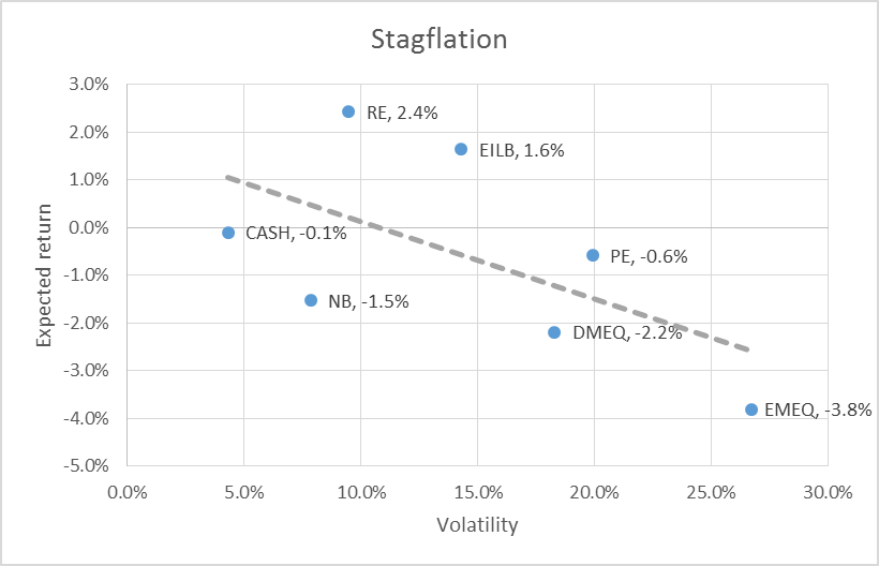
Robust Portfolio Construction with Scenarios

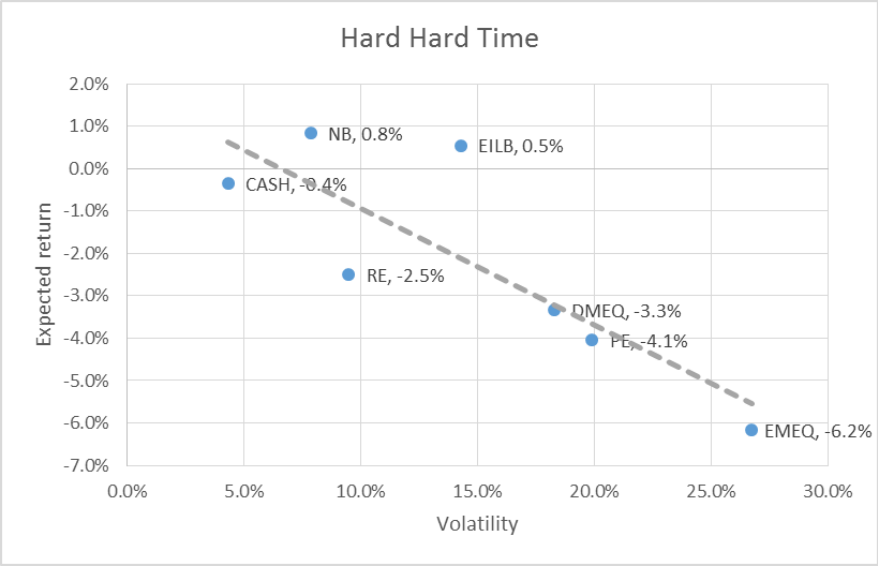


An illustrative example for regret robust optimization









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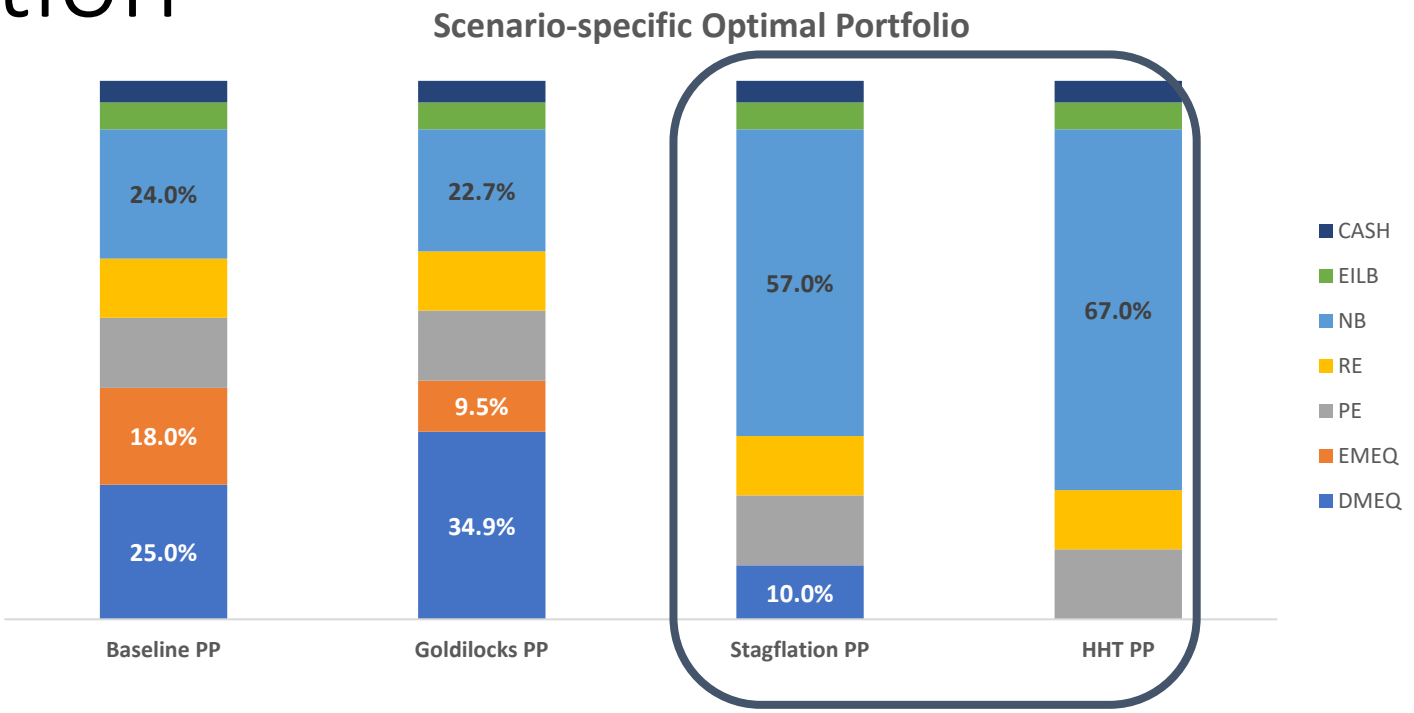

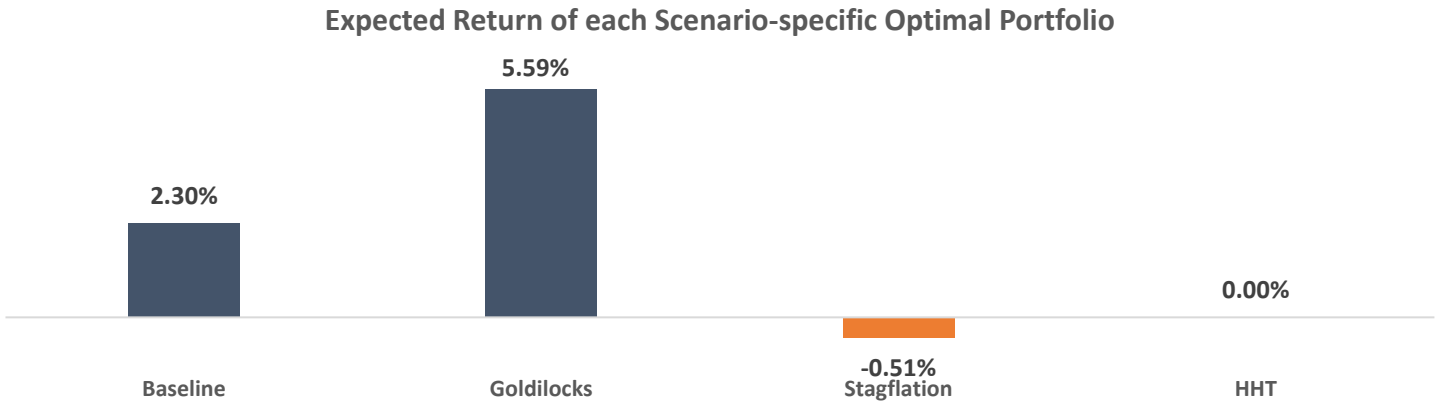

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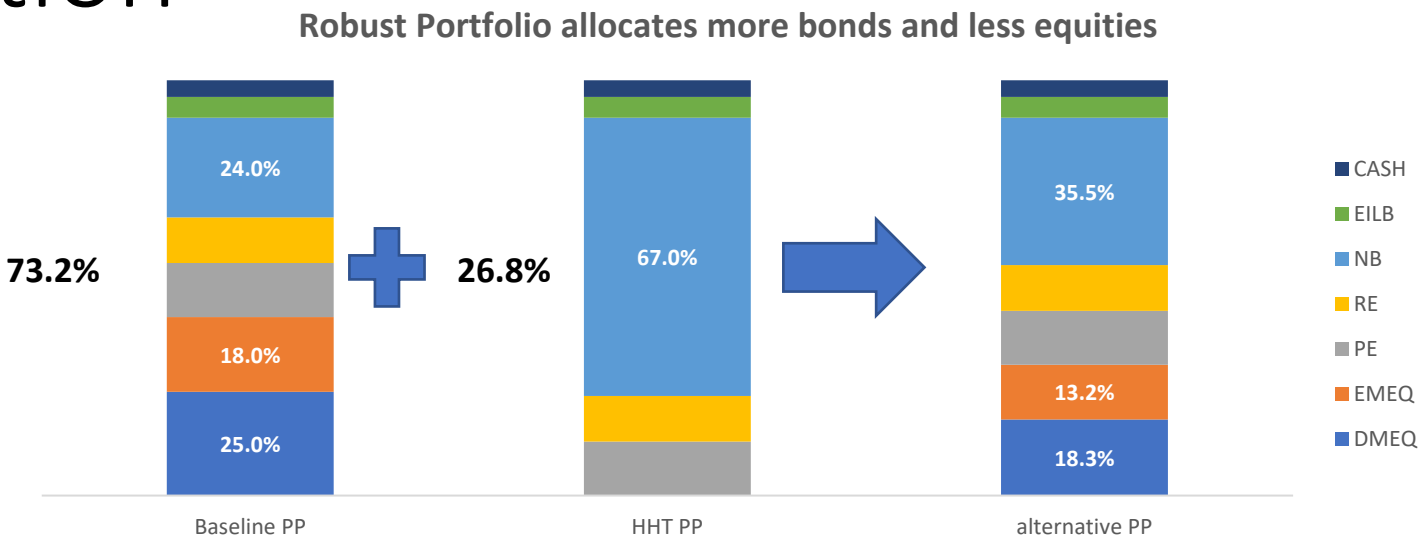
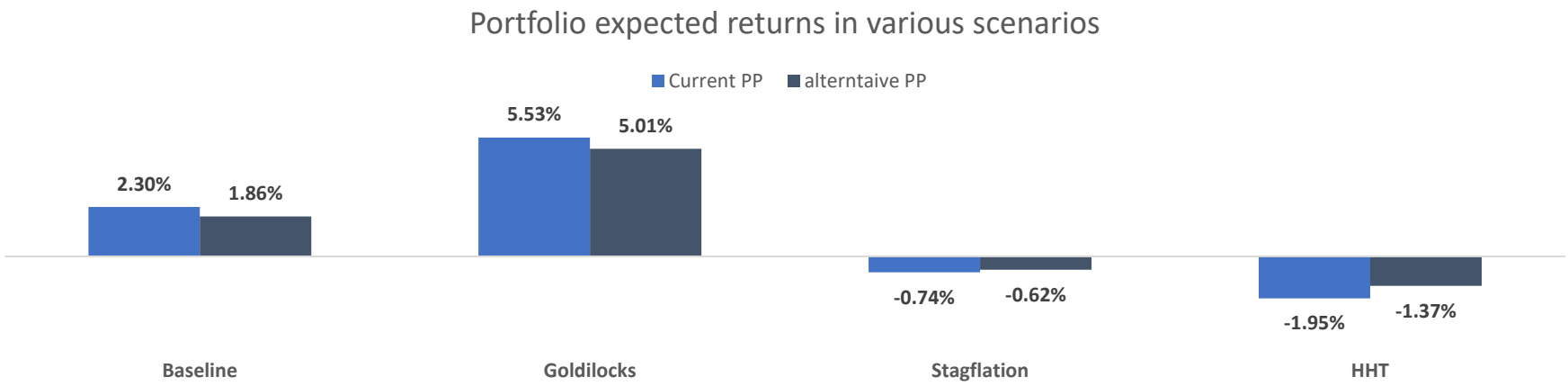
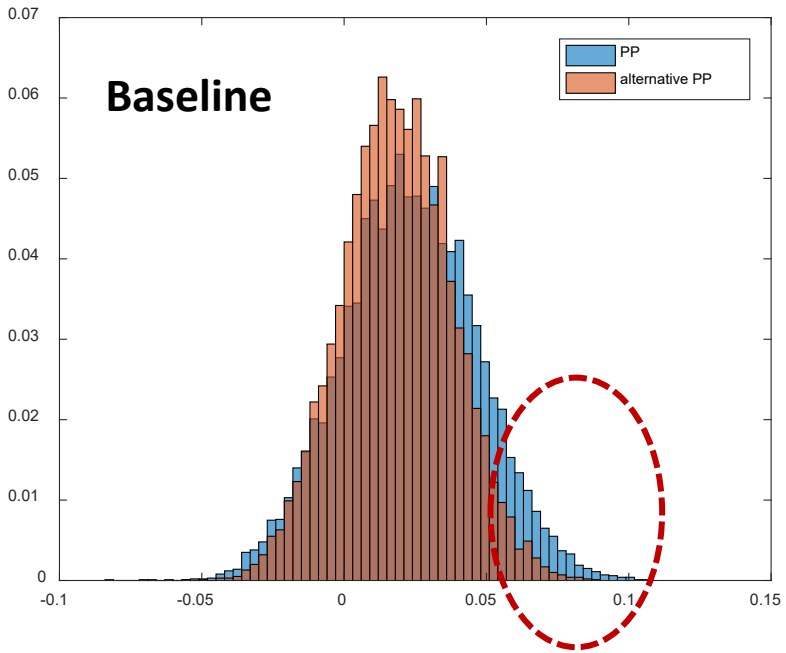
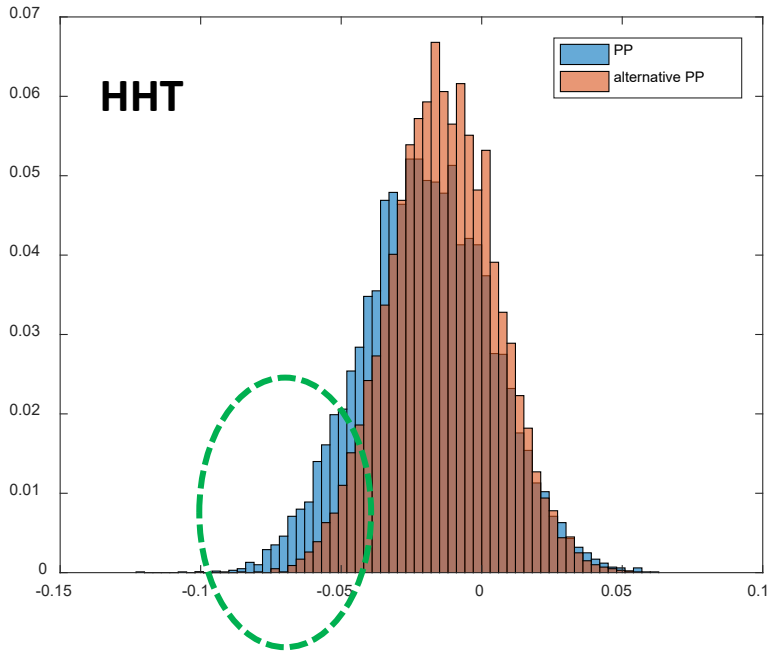


Investment Regret	Baseline	Goldilocks	Stagflation	HHT	Overall Regret
Baseline PP	0.00%	0.06%	0.23%	1.96%	**0.86%**
Goldilocks PP	0.53%	0.00%	0.19%	1.86%	**0.91%**
Stagflation PP	1.66%	1.64%	0.00%	0.34%	**1.37%**
HHT PP	1.87%	2.18%	0.01%	0.00%	**1.57%**
Alternative PP	0.44%	0.58%	0.11%	1.37%	**0.71%**

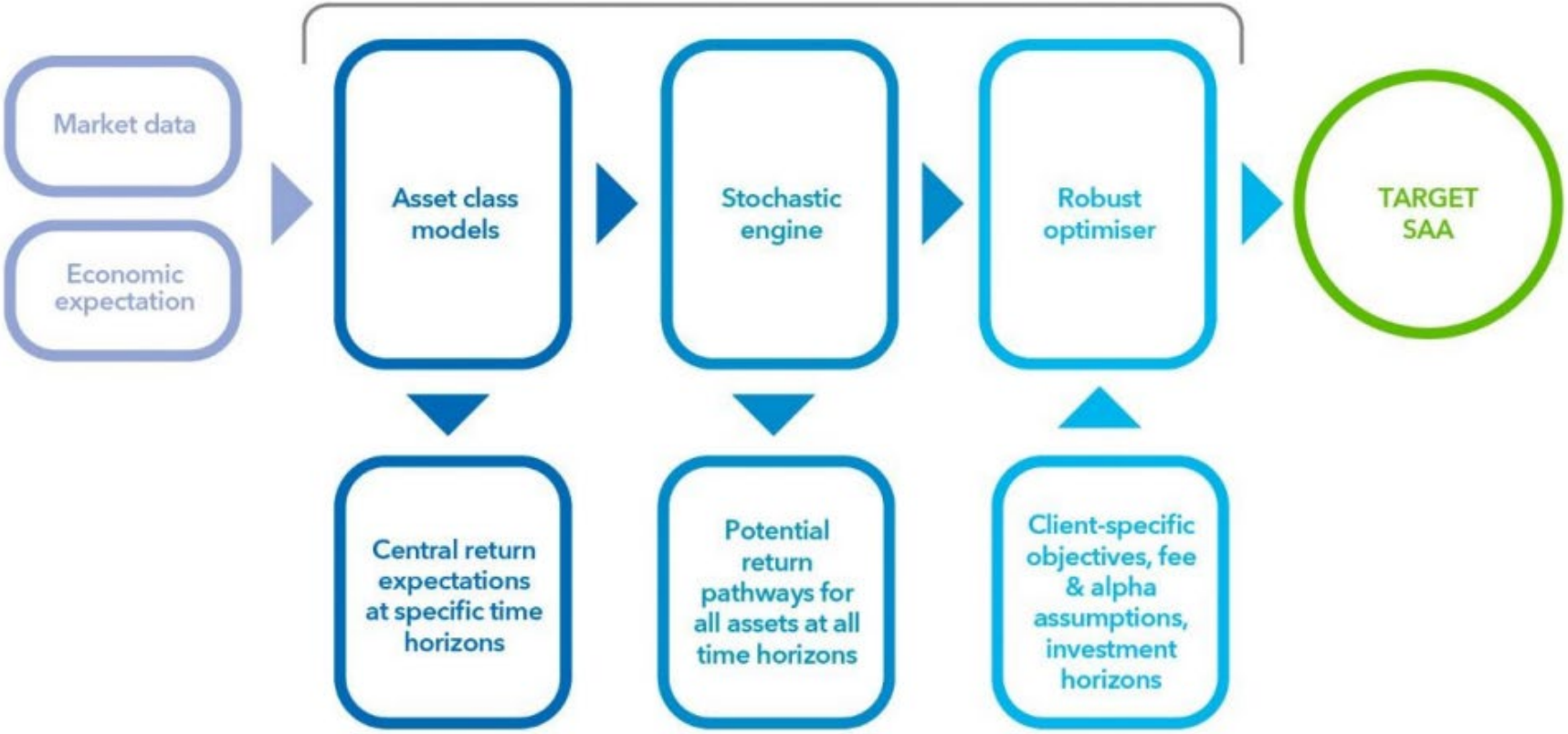
An illustrative example for regret robust optimization







Robust Portfolio Construction with Simulations



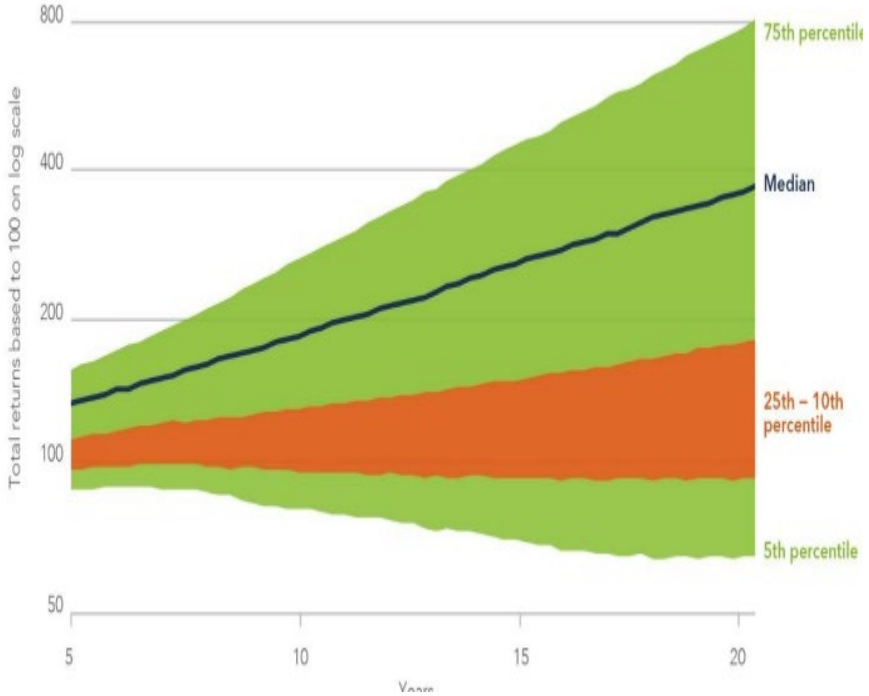
Sources: BlackRock Investment Institute, December 2018. Notes: This graphic is for illustrative purposes only and subject to change without notice.

Robust Portfolio Construction with Simulations

- Let's use 15,000 simulations for purely illustrative purposes. These simulation pathways take into account both variability in expected returns (uncertainty) and historic volatility of the asset class (risk).

- For any given portfolio, we can compute the portfolio's return expectation (or any other metric or objective of interest) for each path giving us 15,000 potential outcomes in this instance. We can sort and pick say the bottom decile, quartile or any other percentile of our choosing and search for the portfolio allocation that **maximizes returns for those outcomes**.

- Choosing the entire set of 15,000 potential outcomes to optimize over will effectively amount to MVO and allocations will tend to exhibit all the symptoms of the approach such as concentrated positions and corner solutions in the absence of subjective constraints or heuristics.



Robust Portfolio Construction Comparison

- The two approaches (scenario-based, simulation-based) incorporating uncertainty as two sides of the same coin. Both aim to address issues with traditional MVO – particularly the over reliance on single point estimates of expected returns and accepting the inherent “unknown unknowns” in investing. Yet there are some differences worth exploring.

- The first difference in approach is philosophical, reflecting different preferences and objectives.
  - The scenario-based approach aims to find the portfolio that is “well-rounded” under different macro scenarios. It minimizes probability weighted regret risk, which in our case study is defined as the return gap from the highest possible return achievable under each scenario. An asset class can be rewarded or penalized under such macro uncertainty, depending on whether it performs better or worse under different scenarios.
  - A simulation-based approach focuses in contrast on portfolio resilience, aiming to find the portfolio that maximizes the expected return in a lower percentile of the asset return simulations. Choosing a smaller percentile of the simulations is analogous to reflecting a higher aversion to uncertainty and a wider confidence band around the capital market assumption. Since this approach does not differentiate between upside and downside uncertainty, an asset class with high uncertainty will be penalized in terms of allocation.

Robust Portfolio Construction Comparison

- The second difference in approach more technical.
  - A scenario-based approach forms a handful of explicit macro scenarios, with separate sets of CMAs underlying each scenario and a probability assigned to each scenario. This approach explicitly accounts for macro uncertainty, but perhaps not other types of asset return uncertainty under each of these scenarios.
  - A simulation-based approach assumes only one single central base case macro scenario. Yet multiple simulations of asset returns are generated to reflect the uncertainty in the asset return estimates due to model error, manager dispersion and other factors, and only implicitly for uncertainty in the macro scenarios.